

Discording Life Balance: A Predictive System for Gaming-Affected Students

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Abstract

This paper presents a comprehensive data science approach to understanding and supporting university students affected by gaming-related lifestyle challenges, with a particular focus on time management and mental health issues. Leveraging three large-scale, survey-based datasets, the study integrates advanced data preprocessing, feature engineering, and supervised learning to develop predictive models that classify students at risk of poor time management or mental health concerns based on behavioral, academic, and psychological indicators.

Classification models, including decision trees and logistic regression, are employed to extract interpretable rules and quantify the influence of factors such as gaming frequency, sleep habits, academic pressure, and psychological stress. To further uncover latent student profiles, K-Modes clustering, optimized via the elbow method and robust initialization, segments the student population into distinct groups, revealing nuanced patterns linking gaming behavior, academic performance, and well-being. Pattern mining using the Apriori algorithm identifies actionable associations, highlighting both expected and counterintuitive relationships among lifestyle factors, academic outcomes, and mental health status.

The findings inform the design of DiTTo, a proposed Discord-based extension bot that delivers personalized recommendations and resources to students, aiming to promote healthier routines and proactive mental health awareness. The report details the methodology, evaluation, and key insights, demonstrating that data-driven interventions can effectively identify at-risk students and tailor support strategies. This work sets the stage for future deployment of intelligent, context-aware academic wellness assistants within student communities.

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1 Introduction

The rapid growth of online gaming and digital entertainment has significantly influenced the lifestyles and daily routines of university students worldwide. While gaming offers recreational value and opportunities for social interaction, excessive or poorly managed gaming habits can negatively impact students' academic performance, time management, and mental health. University students, in particular, often struggle to balance academic responsibilities, social life, and personal well-being, making them vulnerable to issues such as poor time management, sleep deprivation, and increased stress levels.

In recent years, there has been a growing interest in leveraging data-driven approaches to better understand and address these challenges. By analyzing patterns in students' gaming behaviors, academic outcomes, well-being indicators, and support services can develop targeted interventions and tools to help students achieve a healthier balance.

This project aims to design and implement a data science pipeline that preprocesses and analyzes student survey data related to gaming habits, academic performance, and well-being. Our primary goal is to develop a classification system that can identify students at risk of time management or mental health issues based on their self-reported behaviors and experiences. The insights gained from this analysis will support the development of a Discord extension bot, which will provide personalized recommendations and resources to students, thereby promoting better time management and mental health awareness.

The report is organized as follows: Section 2 describes the datasets used in this study. Section 3 outlines the methodologies applied for preprocessing, feature engineering, and classification. Section 4 presents the evaluation of our classification approach. Section 5 discusses the findings, limitations, and potential improvements. Finally, Section 6 details the roles and responsibilities of each team member involved in this project.

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2 Dataset

2.1 Gaming Effect on Students and Academic Result (Parent Dataset)

This dataset, sourced from Kaggle, contains survey responses from university students regarding their gaming habits and academic outcomes. It includes approximately 149 records (students) and features such as:

- **Timestamp:** The time the survey was performed.
- **Age:** The age of the respondents.
- **Gender:** Respondents' gender.
- **Education level:** Respondent educational level (Higher Secondary Certificate(HSC), Undergraduate, Postgraduate).
- **Are you playing online games:** Yes or No answer indicating whether the respondent is playing online game or not.
- **Time spent on gaming:** Self-reported average daily gaming hours.
- **When do you play games:** Preferred gaming time (morning, afternoon, evening, midnight).
- **Sleep impact:** Whether gaming hampers sleep (yes, sometimes, no).
- **Headache:** Self-reported headache (yes, sometimes, no).
- **Mental stress:** Self-reported mental health indicators (yes, sometimes, no).
- **Depression:** Self-reported depression (yes, sometimes, no).
- **Reading attention level after gaming:** Self-reported reading attention (bad, average, good excellent).
- **Academic performance:** GPA or grade information.

This dataset serves as the foundation for identifying initial patterns and constructing labels for time management and mental health issues.

2.2 uniGame Dataset (Time Management Child Dataset)

The uniGame dataset, obtained from HuggingFace, offers a more detailed look at students' academic routines and lifestyle factors. It includes 989 records and the following key features:

- **What is your age:** The age of the respondent.
- **Current educational position:** Respondent educational level.
- **Gender?:** The gender of the respondent.
- **Current GPA:** Academic performance of the respondents.
- **Your Higher Secondary School(H. SC) or A level or equivalent result?:** The higher secondary school result of the respondent.
- **At what age you had started playing games?:** The age at which the respondent started playing games.
- **Do you play games on mobile or pc?:** The device respondent use to play games.
- **When you go to sleep?:** Reported bedtime in 24-hour format.
- **Morning class attendance:** Regularity of attending morning classes.
- **Average gaming time:** Daily hours spent gaming.

- **Which type of game you addicted?:** Type of game respondent addicted to.
- **Do you read newspaper:** How often respondent read newspaper.
- **Do you play paid or non-paid games?:** Whether respondent play paid or non-paid games.
- **Social engagement:** Time spent with family and friends.
- **Respondent feeling if they cannot play game in a day:** Whether respondent feels positive or negative when they cannot play game in a day.
- **Feeling after completing game level:** How respondent feel about completing a game level (excited or bored).
- **Playing game purpose:** Does respondent play games to relieve stress.
- **Fatigue:** Self-reported fatigue (yes/no).
- **Are you wearing glasses?:** Respondent eye condition.

This dataset is primarily used for in-depth time management analysis, classification, clustering and pattern mining.

2.3 OpenML Mental Health Dataset (Mental Health Child Dataset)

The third dataset, accessed via OpenML, focuses on mental health indicators among students. It contains approximately 27901 records and includes:

- **ID:** Respondent identity number (27901 distinct values).
- **Gender:** Respondent gender (female, male).
- **Age:** The age of the respondent.
- **City:** The city where the respondent comes from.
- **Profession:** Respondent profession.
- **Academic Pressure:** Reported academic pressure on scale 0 to 5.
- **Work Pressure:** Reported work pressure on scale 0 to 5.
- **CGPA:** Respondent CGPA.
- **Study Satisfaction:** Reported study satisfaction on scale 0 to 5.
- **Job Satisfaction:** Reported job satisfaction on scale 1 to 5.
- **Sleeping Duration:** Respondent daily average sleeping time.
- **Dietary Habits:** Respondent eating habit (unhealthy, moderate, healthy).
- **Degree:** Respondent degree major.
- **Suicidal Thoughts:** Whether respondent have thoughts of suicide before.
- **Work Study Hours:** The hours respondent spend daily for work and study.
- **Financial Stress:** Indicating whether respondent has financial stress (yes or no).
- **Family History:** Reported depression history among family members.
- **Depression:** 1 or 0 indicates respondent depression state.

This dataset supports the classification and analysis of mental health issues, complementing the information from the parent dataset.

3 Methodologies

3.1 Preprocessing

Before any modeling or pattern discovery could be performed, all three datasets underwent a thorough preprocessing phase. The original datasets contained a mix of numerical and categorical data, as well as attributes that were either irrelevant or redundant for our analysis. The preprocessing steps included removing unnecessary attributes, handling missing or inconsistent data, grouping raw values into meaningful categories, assigning ordinal scores to these categories, and engineering labels for supervised learning. This process not only improved data quality but also ensured that the resulting features were interpretable and suitable for classification, clustering, and pattern mining tasks. The use of binning (discretization), categorical encoding, and scoring is widely recognized in data mining and machine learning literature as a best practice for transforming raw data into features suitable for predictive modeling and pattern analysis [2, 7, 9].

3.1.1 Parent Dataset.

For the Gaming Effect on Students and Academic Result dataset, we focused on five key attributes most relevant to time management and mental health. Each attribute was transformed from its raw form into categorical ranges, then encoded with a risk-based score.

The attribute “how much time is spent on gaming” was divided into three categories: less than 2 hours was considered low and assigned a score of 0, 3–4 hours was categorized as average with a score of 0.5, and more than 5 hours was considered high, receiving a score of 1. This reflects the increasing risk of poor time management as gaming time increases. The attribute “when do you play games the most” was also categorized, with Morning and Afternoon both scored as 0, Evening as 0.5, and Midnight as 1, since later play times are more likely to disrupt daily routines and sleep. For “do you feel hamper in sleep,” the responses were mapped to Yes (1), Sometimes (0.5), and No (0), capturing the degree to which gaming affects sleep quality.

The mental health-related attributes, “do you feel mental stress” and “do you feel depression,” were both scored in the same way: Yes (1), Sometimes (0.5), and No (0). This approach allows us to quantify the severity of mental health concerns. For classification, we grouped the first three attributes (gaming time, play time, and sleep impact) as leading to time management issues, while the last two (mental stress and depression) were used for mental health classification. To create a seriousness score, we summed the relevant attribute scores and applied a weight (1.0 for time management, 1.5 for mental health). Students whose seriousness score exceeded a set threshold were labeled as needing intervention. This scoring and thresholding approach was necessary because the dataset did not provide explicit labels for these outcomes. The rationale for creating composite scores and thresholds is supported by standard practices in feature engineering and supervised learning, as described by [2, 7].

3.1.2 Time Management.

The uniGame dataset originally included 20 attributes, but was reduced to six core features directly related to student time management. Each attribute was carefully binned or recoded to enhance interpretability and analytic value.

The GPA attribute, which ranged from 2.00 to 4.00, was binned into three groups using quantile binning: Low (2.00–3.00), Average (3.00–3.41), and High (3.41–4.00), and assigned scores of 0, 1, and 2, respectively, with lower GPA indicating higher risk. The “when you go to sleep” attribute, with ten distinct values, was grouped into four categories: Healthy (20–22, scored 0), Average (23, 0, 1, scored 1), Late (2–4, scored 2), and Very Late (5–8, scored 3), reflecting the impact of late sleep on time management. “Attending morning class regularly” was treated as a binary variable: Yes (0) and No (1), with non-attendance indicating higher risk. The “average time spent on playing games” attribute was categorized as Low (0–2 hours, 0), Moderate (3–5 hours, 1), and High (6+ hours, 2), mirroring the structure used in the parent dataset. “Time spent with family and friends” was inversely scored, as higher social time is protective: Low (0–2 hours, 2), Moderate (3–5 hours, 1), and High (6+ hours, 0). Finally, “do you feel fatigue” was binary: Yes (1) and No (0).

The total time management score for each student was calculated by summing the scores across all six attributes. Students with a total score from 0 to 5 were labeled as having no time management issue, while those scoring 6 to 12 were flagged as needing intervention. This scoring system allowed us to generate labels for supervised classification, which are essential for training decision tree models using the Gini index. The process of quantile binning and ordinal encoding is a recognized and recommended technique for handling continuous and ordinal data in predictive modeling [2, 6].

3.1.3 Mental Health.

The mental health dataset included ten attributes, each preprocessed to reflect its contribution to mental health risk. GPA was quantile-binned into Low (5.00–6.89), Average (6.90–8.56), and High (8.57–10.00), aligning with the approach used in the time management dataset. Academic pressure and study satisfaction, both on a 1–5 scale, were grouped as Low (0–2), Moderate (3), and High (4–5). Sleep duration was categorized as Very Short (<5 hours), Short (5–6 hours), Adequate (7–8 hours), and Long (>8 hours), with shorter sleep indicating higher risk. Dietary habit was classified as Unhealthy, Moderate, or Healthy. Suicidal thoughts, family history, and depression were all binary attributes (Yes = 1, No = 0). Work/study hours were grouped as Low (0–4 hours), Average (5–8 hours), and High (9–12 hours). Financial stress was quantile-binned into Low (≤ 2), Moderate (3–4), and High (5).

For mental health classification, a composite risk score was calculated by summing the scores across all attributes. Students exceeding a threshold (determined by score distribution analysis) were labeled as needing mental health support. This approach enabled us to create meaningful labels for supervised learning, even in the absence of explicit outcome variables in the raw data. The methodology of engineering labels from composite scores and using them for supervised machine learning is described in [2, 9].

3.2 Classification

3.2.1 Parent Dataset.

The methodology for the Parent Classification task focused on identifying whether a student's primary concern lies in time management or mental health. The dataset contained lifestyle survey responses categorized into two domains: time management (gaming time, gaming frequency) and mental health (stress, headache, sleep). Since all input variables were categorical (e.g., "Yes", "No", "Sometimes"), they were first converted into numeric scores: "Yes" was assigned 1, "No" as 0, and intermediate responses like "Sometimes" as 0.5. Each domain's score was averaged separately. To account for the imbalance in question count (2 time management vs. 3 mental health), the time management score was multiplied by 1.5, and the mental health score by 1.0 to normalize contributions. The final label was determined based on which weighted score was higher.

A Decision Tree Classifier was trained to extract interpretable decision rules, using the Gini index to determine the optimal splits at each node. This splitting criterion is commonly used in tree-based classifiers to reduce impurity and improve predictive accuracy [8]. In parallel, a Logistic Regression model was trained to provide a linear view of feature importance. Both models were evaluated using metrics such as accuracy, precision, recall, and F1-score, and they confirmed that behaviors like late-night gaming and frequent stress are indicative of mental health challenges, even in high-performing students.

3.2.2 Time Management.

The Time Management Classification used behavioral and lifestyle data from the uniGame dataset to identify students at risk of poor time management. The dataset included features such as GPA category, sleep habits, gaming hours, morning class attendance, fatigue, and social interaction. All categorical features were mapped to numeric risk scores: 0 for low-risk/healthy behavior, 1 for moderate concern, and 2 or 3 for high-risk behavior (e.g., gaming over 6 hours/day or minimal social time). All features were treated equally, and no weighting was applied. A total risk score was computed for each student, and students with scores 6 were labeled as "Yes" (needs time management support), while those scoring ≤ 5 were labeled "No".

A Decision Tree Classifier was applied to uncover branching decision logic, where the Gini index was again used to evaluate the best feature splits [8, 10]. This tree provided insight into key indicators such as fatigue, skipping classes, and high gaming frequency. A Logistic Regression model was also trained to understand linear associations, highlighting that time spent gaming and lack of social time were significant predictors. Both models confirmed that poor time routines could be detected early, even in students with good academic performance.

3.2.3 Mental Health.

The Mental Health Classification is used to predict whether a student was experiencing depression, as indicated by the Depression label in the dataset. The dataset included over 27,000 records

with features related to academic performance, lifestyle, and psychological factors. These included CGPA (categorized), academic pressure, study satisfaction, sleep duration, dietary habits, suicidal thoughts, work-study hours, financial stress, and family history. All categorical features were transformed into numerical scores based on severity (e.g., "Very Short" sleep = 0, "Healthy" diet = 2). No weighting was applied, as each feature was considered to contribute equally to mental health risk, and rows with missing values were removed to ensure clean data. Students were labeled as "Yes" if they were identified as experiencing depression, and "No" otherwise.

A Decision Tree Classifier was trained with a maximum depth of 3 and `class_weight='balanced'`, a technique that compensates for label imbalance by increasing the model's sensitivity to underrepresented classes [1]. The model used the Gini index to determine effective split criteria [8, 10]. It found that suicidal thoughts, financial stress, and academic pressure were the most influential predictors. In addition, a Logistic Regression model was trained for a linear validation of the decision rules. The regression coefficients showed the same trends: suicidal ideation and stress had strong positive associations with depression, while features like study satisfaction, good sleep, and healthy eating had negative effects. Both models showed high recall and precision, emphasizing that mental health is often driven by an accumulation of moderate risks rather than a single severe factor.

3.3 Clustering

3.3.1 K-Modes Clustering.

K-Modes clustering represents a significant extension of the traditional K-Means algorithm, specifically designed to address the limitations inherent in clustering datasets dominated by categorical attributes [3]. The motivation for developing K-Modes stemmed from the computational inefficiency of standard hierarchical clustering methods for large datasets and the restriction of K-Means-based approaches to numeric data only. While K-Means clustering operates on numerical data using Euclidean distance measures and arithmetic means to represent cluster centroids, K-Modes fundamentally differs by employing modes (the most frequently occurring values) as cluster representatives and utilizing discrete dissimilarity measures appropriate for categorical data. This algorithmic adaptation makes K-Modes particularly well-suited for datasets with predominantly categorical attributes, as it naturally handles the non-ordered nature of categorical domains without requiring artificial numeric conversions that may produce meaningless results. The algorithm preserves the computational efficiency of K-Means with a time complexity of $O((t+1)kn)$, where t represents the number of iterations, k the number of clusters, and n the number of objects, making it highly scalable for large-scale data mining applications.

The technical foundation of K-Modes clustering relies on the Hamming distance as the primary dissimilarity measure [5], which quantifies the number of categorical attributes that differ between two data objects. Mathematically, the dissimilarity between two categorical data objects X and Y with m attributes is expressed as

$$d(X, Y) = \sum_{j=1}^m \delta(x_j, y_j) \quad (1)$$

where

$$\delta(x_j, y_j) = \begin{cases} 0 & \text{if } x_j = y_j \\ 1 & \text{if } x_j \neq y_j \end{cases}$$

This basic dissimilarity measure gives equal importance to each category of an attribute. To account for the frequencies of categories in a dataset, an alternative chi-square distance can be employed where the chi-square distance gives more importance to rare categories than frequent ones and proves useful in discovering under-represented object clusters.

The algorithm[4] operates through an iterative process beginning with the random selection of k initial cluster prototypes, followed by the assignment of each data object to the cluster whose prototype minimizes the dissimilarity measure. Subsequently, cluster prototypes are updated by determining the mode of each categorical attribute within the respective clusters, effectively representing each cluster by its most characteristic categorical values. This iterative assignment and update process continues until convergence is achieved, typically when no objects change cluster membership between consecutive iterations.

3.3.2 Initialization Methods for K-Modes Clustering.

The initialization of cluster prototypes critically influences the performance and convergence behavior of the K-Modes algorithm, as poor initial choices often lead to suboptimal local minima solutions. Mathematically, the algorithm minimizes the within-cluster dissimilarity cost function

$$P(W, Q) = \sum_{l=1}^k \sum_{i=1}^n w_{i,l} d(X_i, Q_l) \quad (2)$$

where d is the Hamming distance for categorical attributes. Huang (1998) demonstrated that convergence speed and solution quality depend heavily on initial modes Q , as random initialization often requires multiple restarts to avoid high-cost local optima. This occurs because the iterative assignment-update process, where objects are assigned to the nearest mode, and modes are updated to the most frequent category per attribute is non-convex and highly sensitive to starting points. Consequently, refined initialization methods are essential to reduce iteration counts and improve clustering accuracy.

Two prominent initialization strategies are Huang's methods and Cao's density-based approach. Huang proposed two techniques:

- (1) **First-k Distinct Objects:** Selects the first k unique records from the dataset. While computationally efficient, this method risks bias if initial records are unrepresentative.
- (2) **Frequency-Based Assignment:** Assigns the most frequent attribute categories uniformly across k initial modes, then replaces them with the closest data objects. This promotes diversity but lacks a rigorous criterion for handling ties or rare categories.

In contrast, Cao's method defines an object's average density, which quantifies local concentration based on attribute-value frequencies. The first cluster center is the object with maximal density. Subsequent centers are chosen by maximizing $d(x, xm)\text{Dens}(x)$, where d is the Hamming distance to existing centers xm . This dual

emphasis on density (avoiding outliers) and distance (ensuring dispersion) yields more robust initial prototypes. Huang's methods are simple, efficient ($O(k)$) but risk poor representation in skewed datasets and are unstable under data reordering. While Cao's approach demonstrates superior robustness and achieves higher accuracy (8-12% gains in precision across UCI datasets) by anchoring centers in high-density regions while maintaining dispersion, it comes at a cost of higher computational cost. Consequently, Cao's method was implemented in this project due to its robustness to initialization variance and consistent convergence to lower-cost solutions.

3.3.3 Elbow Method for Finding Optimal K-value.

Determining the optimal number of clusters (K) is crucial in K-Modes clustering, as selecting an inappropriate K value can lead to either oversimplified clustering (underfitting) or excessive fragmentation (overfitting), ultimately compromising the interpretability and quality of the clustering results. The elbow method provides a systematic approach to identify the optimal K by plotting the clustering cost function against different values of K and identifying the point where the rate of cost reduction significantly diminishes. In K-Modes clustering, this involves iteratively running the algorithm with varying cluster numbers (typically from 1 to n) and recording the total within-cluster dissimilarity cost for each K value.

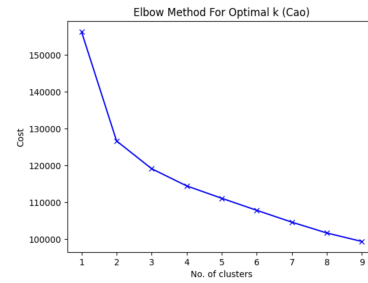


Figure 1: Elbow Method

```
# Elbow curve to find optimal K (Cao)
cost_cao = []
K = range(1,10)

for num_clusters in list(K):
    kmode_cao = Kmodes(n_clusters=num_clusters, init = "Cao", n_init = 5, verbose=1)
    kmode_cao.fit_predict(global_feature)
    cost_cao.append(kmode_cao.cost_)

plt.plot(K, cost_cao, 'bo-')
plt.xlabel('No. of clusters')
plt.ylabel('Cost')
plt.title('Elbow Method For Optimal k (Cao)')
plt.show()
```

Figure 2: Elbow Method Code

The resulting elbow plot visualizes the trade-off between cluster granularity and within-cluster homogeneity, where the "elbow" represents the point of diminishing returns. As demonstrated in the provided example, the cost function decreases sharply from $K=1$ to $K=3$, after which the rate of decrease becomes more gradual, suggesting that $K=3$ represents the optimal number of clusters for this particular dataset. This inflection point indicates that adding more clusters beyond $K=3$ provides minimal improvement in clustering quality while increasing computational complexity and potentially introducing noise.

3.3.4 K-Modes Clustering on Datasets.

The clustering analysis was conducted on three interconnected datasets, each capturing distinct but related aspects of student behavior and academic performance. The parent dataset, "Gaming Effect on Students and Academic Result," serves as the primary dataset encompassing 9 categorical attributes that comprehensively profiles the relationship between gaming habits, health-related factors, and academic outcomes.

The first child dataset, "Time Management Dataset," focuses on disciplinary and time allocation patterns with attributes that provide insights into how students manage their balance of daily schedule and academic activities, allowing us to investigate the potential correlation between time management practices and academic performance. On the other hand, the second child dataset, "Mental Health Dataset," is designed to investigate the complex interplay between students' academic standing, psychological well-being, and health-related behaviors. By capturing categorical indicators of academic performance, mental stress, lifestyle habits, and psychological risk factors, this dataset enables the identification of stress patterns, factors and subgroups within the student population.

The clustering process follows a standardized methodology applied consistently across all three datasets, beginning with data preprocessing and DataFrame conversion to ensure compatibility with the K-Modes algorithm. The implementation, as described in Figure 3, utilizes the Python *kmodes* library, with the process initiated by loading the dataset and additional data frame processing to ease the clustering operations.

The first critical step involves applying the elbow method to determine the optimal number of clusters, which varies across datasets. This process iteratively tests different K values and plots the cost function to identify the point of diminishing returns. Following K-value optimization, the actual clustering is performed using the Cao initialization method, with the fitted model applied to the dataset. The cluster centroids are then extracted and converted to a DataFrame format for analysis, with column names mapped to match the original feature names.

The clustering results are then integrated with the original dataset by appending cluster assignments, creating a comprehensive dataset that combines original attributes with cluster membership for subsequent analysis and visualization. This systematic approach ensures reproducibility and enables clustering result analysis across all three datasets, facilitating the identification of consistent patterns and relationships between gaming behavior, time management, mental health, and academic performance.

3.4 Pattern Mining

To support the decision-making logic behind our model, we applied pattern mining to uncover interpretable and actionable patterns from behavioral data. Specifically, we used association rule mining with the Apriori algorithm, which is well-suited for extracting "if-then" relationships from categorical survey data [11]. This method enables us to translate user-provided inputs into explainable insights, making it a natural fit for our chatbot, which must justify its recommendations to users. The primary goal was not only to detect

```

1 # Required packages
2 import numpy as np
3 import pandas as pd
4
5 # Import kmodes as chosen Clustering method
6 !pip install kmodes
7 from kmodes.kmodes import KModes
8
9 # Import dataset file (CSV)
10 df = pd.read_csv('dataset/student.csv')
11 global_feature = df.copy()
12
13 # 1. Conduct Elbow Method search to find optimal K (Cao) - refer Elbow Method section.
14
15 # 2. K-Modes clustering with Cao Initialization, n-clusters (K) = 3
16 km_cao = KModes(n_clusters=3, init = "Cao", n_init = 1, verbose=1)
17 global_fitcluster_cao = km_cao.fit_predict(global_feature)
18
19 # 3. View cluster centroids
20 global_cluster_centroids = pd.DataFrame(km_cao.cluster_centroids_)
21 global_cluster_centroids.columns = global_feature.columns
22 global_cluster_centroids
23
24 # 4. Turn clustering results (global_fitcluster_cao) into dataframe
25 global_fitcluster_cao_df = pd.DataFrame(global_fitcluster_cao)
26 global_fitcluster_cao_df.columns = ['cluster']
27
28 # 5. Append predicted clustering results to original dataframe
29 global_feature_clustered = global_feature.copy()
30 global_feature_clustered['cluster'] = global_fitcluster_cao_df
31 global_feature_clustered
32
33 # 6. Proceed with clustering result visualization/analysis

```

Figure 3: K-Modes Clustering Example Code

individual risk factors, but to reveal combinations of behaviors that frequently co-occur with depression or poor academic performance. This approach ensures the bot remains transparent, explainable, and adaptable to a variety of behavioral profiles [12].

3.4.1 Time Management.

In the time management and academic performance dataset, our objective was to identify lifestyle behaviors that correlate with different CGPA categories (e.g., High, Average, Low). Students reported their habits such as gaming frequency, sleep patterns, study hours, and time spent with family and friends. As with the previous dataset, we applied Apriori to derive rules where the consequent was a single CGPA class. We retained only rules with at least two antecedent items to emphasize richer behavioral combinations. For example, we observed that Gaming=High and Sleep=Late often co-occur with CGPA=Low, while more balanced habits like Sleep=Average and Family Time=Moderate are linked to CGPA=High. These findings help shape the model's coaching features, allowing the bot to provide tailored feedback on academic routines and suggest adjustments based on user patterns. [12]

3.4.2 Mental Health.

For the depression-focused dataset, we analyzed categorical features such as academic pressure, suicidal thoughts, sleep duration, fatigue, and lifestyle habits. Pattern mining was particularly appropriate for this case because it produces interpretable rules—for instance: "If a student reports high academic pressure and suicidal thoughts, then they are likely to be depressed." These multi-condition rules enable the model to detect mental health risks from user responses and offer early intervention. We filtered for rules where the consequent was Depression=1 or Depression=0, focusing on high-confidence rules with at least two conditions in the antecedent. While some common-sense patterns are easy to detect, this project becomes especially valuable when discovering less obvious, rare patterns that may not be widely known but still reflect emerging behavioral risks.

4 Evaluation

4.1 Parent Dataset

4.1.1 Classification.

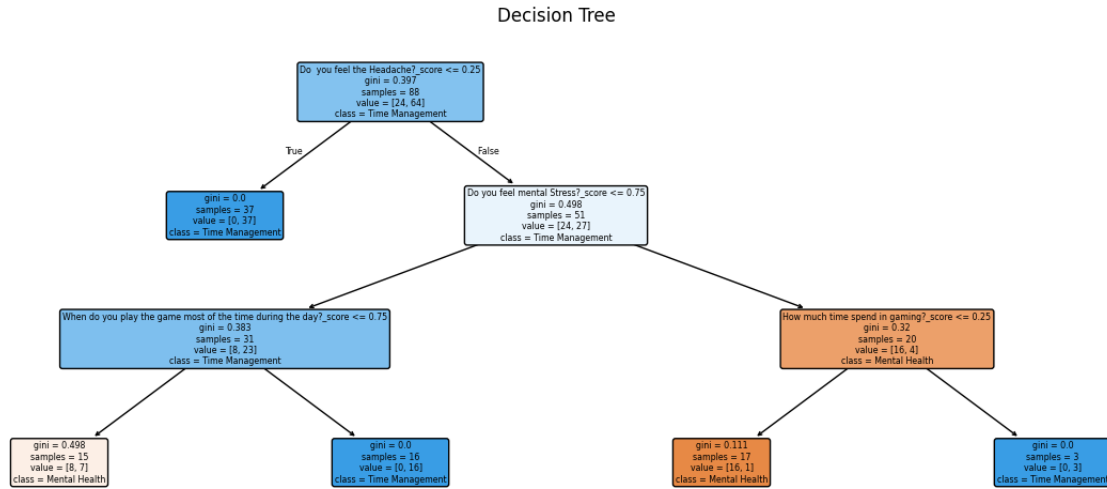


Figure 4: Decision Tree for Classifying Issue in Parent Dataset

Based on Figure 4, the root of the tree splits on the feature “**Do you feel the Headache?**” (score ≤ 0.25), identifying it as the most informative question. Students who report not feeling headaches (score ≤ 0.25) are directed to the left subtree, where all 37 students are classified as having Time Management issues, with perfect purity (Gini = 0.0). This indicates that a lack of physical symptoms strongly correlates with non-mental-health-related struggles.

The right branch of the tree captures those with higher headache scores (score > 0.25), suggesting a higher probability of mental health concerns. This subtree further splits on:

- **Mental Stress Score** (≤ 0.75): Students with moderate stress are directed to further behavioral indicators like gaming frequency and time of day.
- **When do you play games most during the day?**: This identifies lifestyle patterns, where gaming at irregular or late hours correlates more with Time Management issues, whereas other behaviors tilt toward Mental Health concerns.
- **How much time is spent gaming?**: This decision boundary helps refine classification by evaluating excessive engagement in gaming as a coping mechanism — often linked to Mental Health issues.

Across the leaf nodes:

- A leaf with Gini = 0.0 (16 students) shows perfect classification as Time Management.
- A mixed node (Gini = 0.498) shows a less confident split (8 Mental Health, 7 Time Management).
- The node with 17 samples (Gini = 0.111) is mostly Mental Health, showing low impurity.

Limitations.

- Some leaf nodes show mixed classification (e.g., Gini ≈ 0.5), indicating ambiguity in those branches.
- The model might overfit the data due to small sample size in some branches (e.g., 3–8 students).

- Survey-based self-reported scores may introduce bias or inconsistency.

Implications.

- Physical symptoms like headaches and emotional stress appear as strong early indicators of Mental Health issues.
- Behavioral patterns (such as time and duration of gaming) are more indicative of Time Management problems.

4.1.2 Clustering.

Clustering analysis was performed on the parent dataset using the elbow method, which indicated that three clusters ($K = 3$) provided the optimal grouping. The representative centroids for each cluster are summarized in Figure 6. These centroids serve as the prototypical profiles for each cluster and facilitate interpretation of the clustering structure.

The distribution of clustering results across key categorical variables is visualized in Figure 7 indicates cluster characteristics as follows:

- **Cluster 0**: Primarily consists of students who report no depression, maintain good academic performance, and exhibit average to good attention levels.
- **Cluster 1**: Characterized by students who sometimes experience depression and display a mix of average and good academic and attention outcomes, suggesting an intermediate profile.
- **Cluster 2**: Dominated by students who frequently report depression, lower attention, and comparatively poorer academic performance, indicating a higher-risk group.

Further analysis of gaming-related attributes, as depicted in Figure 8, reveals that Cluster 0 students are regular gamers who predominantly play in the evening or afternoon and report healthier gaming habits. Clusters 1 and 2, however, show a tendency toward late-night (midnight) gaming patterns, with Cluster 2 in particular

exhibiting both problematic gaming behaviors and higher levels of depressive symptoms. This association suggests a potential link between late-night gaming and adverse mental health outcomes.

Physical health-related attributes, including sleep, headache, and stress, are also distinctly distributed across clusters, as shown in Figure 9. Cluster 0 represents students with minimal health issues, Cluster 1 includes those with intermittent problems, and Cluster 2 comprises high-risk students with significant health challenges across all metrics.

Overall, the clustering results successfully identify three distinct student profiles: healthy, moderately affected, and high-risk, with the latter group showing combined academic, mental health, and behavioral concerns.

4.2 Time Management

4.2.1 Classification.

Figure 10 shows the root of the tree begins with **GPA (categorical)**, split at 1.5. This reveals that academic performance is the most influential feature. Students with lower GPA categories (≤ 1.5) are more likely to struggle with time management, while higher GPA students (above 1.5) are generally better at managing their time.

Left Subtree: Students with GPA ≤ 1.5 . The second split evaluates **morning class attendance**. Students who rarely attend morning classes (score ≤ 0.5) are mostly predicted to have time management issues. This branch continues with another split based on GPA (≤ 0.5), providing a finer distinction. Notably: One leaf node shows pure classification (Gini = 0.0) for students with the worst GPA and poor attendance — strong indicators of poor time management. Another node with Gini = 0.414 shows a mixed group. For students with better attendance, the tree considers **time spent with family/friends**. Higher social interaction scores (≤ 1.5) correspond to better time use, but there is still variation: One path classifies students as “No” with Gini = 0.419. Another identifies a more balanced group (Gini = 0.359), classified as “Yes” for good time management.

Right Subtree: Students with GPA > 1.5 . Here, the key split is based on **average time spent playing games**. Students who spend less time gaming (≤ 0.5) are likely to manage their time better — 173 out of 229 samples were classified as “Yes” (i.e., good time management). This subtree further explores: **Fatigue level** (≤ 0.5): Lower fatigue aligns with time-efficient students. One node has very low Gini (0.173) indicating high prediction confidence. **Sleep time**: Students who go to sleep earlier (≤ 0.5) are overwhelmingly classified as good time managers (Gini = 0.111).

Insights and Implications.

- Academic performance and morning class attendance are strong predictors of time management behavior.
- Excessive gaming, fatigue, and poor sleep habits are red flags for time management issues.
- Students with good GPA, consistent attendance, social engagement, and low fatigue are highly likely to exhibit effective time management.

Limitations.

- Some leaf nodes show moderate impurity (Gini ≈ 0.4), suggesting the need for deeper trees or ensemble methods.
- Classification might be sensitive to self-reporting errors in behavioral questions (e.g., fatigue, gaming).

4.2.2 Clustering.

Clustering analysis of the Time Management dataset was conducted using $K = 4$ clusters, as determined by the elbow method. The cluster centroids, summarized in Figure 11, reveal distinct student profiles based on CGPA category, bed time, morning class attendance, average game time, social time, and fatigue.

- **Cluster 0:** Labeled as ‘Average Performing Students’, is characterized by average academic performance, moderate bedtimes, consistent morning class attendance, and balanced engagement in gaming and social activities. This group also reports experiencing fatigue, suggesting a lifestyle that, while balanced, may be somewhat taxing (Figure 12).
- **Cluster 1:** Representing ‘Low Performer Leisure Oriented Students’, is distinguished by low CGPA, late bedtimes, poor morning class attendance, and a preference for leisure activities, with low levels of fatigue. This indicates that lifestyle choices such as late sleeping and reduced academic engagement are associated with poorer academic outcomes.
- **Cluster 2:** Identified as ‘Elite Gamer Scholars’, comprises students who achieve high academic performance despite substantial engagement in gaming activities and moderate social time. These students maintain average bedtimes, do not regularly attend morning classes, and report low fatigue, suggesting an ability to balance gaming with academic responsibilities (Figure 12 14).
- **Cluster 3:** The ‘High Achieving Focused Student’ group, dominates the high CGPA category and is marked by healthy bedtimes, regular morning class attendance, minimal gaming and social time, and low fatigue. This cluster exemplifies disciplined time management and a strong academic focus.

The distribution of clustering results across categorical attributes, as depicted in Figures 12 to 14, highlight clear distinctions in academic performance, lifestyle habits, and fatigue levels among the clusters. Notably, the analysis reveals that factors indicative of strong or poor time management are closely aligned with academic performance, with the exception of a small but distinct group of students who successfully balance high academic achievement with substantial gaming activity. These findings highlighted the impact of time management behaviors on academic outcomes and provide actionable insights for targeted interventions.

4.2.3 Pattern Mining.

As the pattern mining outputs are mostly predictable and not surprising, it would be very beneficial if the unexpected patterns were also given focus. Below are the examples of abnormal patterns that are derived from the time management dataset.

• Fatigue and Sleep vs. Low CGPA

A notable pattern shows that students who do not feel fatigue and report average sleep times are still associated with low CGPA (support 1.6%, confidence = 1.0, lift 3.01, around



Figure 5: Elbow Method to Determine Optimal K for Datasets: Parent, Time Management, Mental Health

Out [47]:

	Playing Games	Game Duration	Game Time	Hamper Sleep	Headache	Stress	Depression	Attention Level	Academics
0	Yes	<2 hours	Mid-Night	No	No	No	No	Average	Good
1	Yes	<2 hours	Evening	Sometimes	Sometimes	Sometimes	Sometimes	Good	Good
2	Yes	<2 hours	Mid-Night	Yes	Yes	Yes	Yes	Average	Good

Figure 6: Cluster Centroids (K = 3) for Parent Dataset.

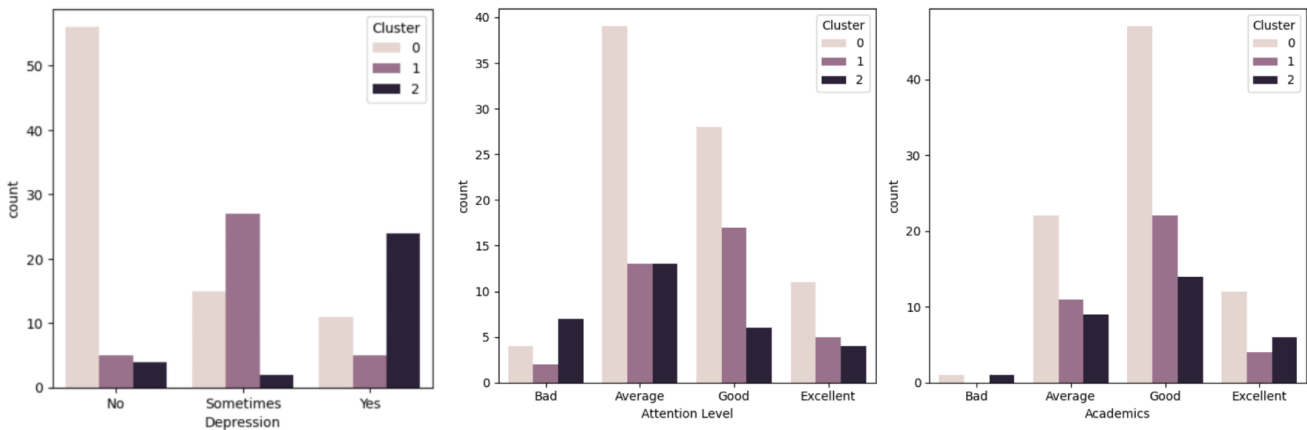


Figure 7: Distribution of Clustering Results Across Categorical Attributes (Parent Dataset: Depression, Attention Level, Academics).

16 students). This is counterintuitive, as one might expect low fatigue and decent sleep to correlate with good academic outcomes. However, it suggests that despite feeling physically fine, these students may lack effective academic engagement or discipline, highlighting the complexity of academic success beyond just restfulness.

- **Skipping Morning Classes and High CGPA**
Another surprising trend indicates that students who do not attend morning classes regularly can still achieve a high

CGPA, with confidence = 1.0 across multiple CGPA thresholds (e.g., 3.5, 3.85, 4.0). This challenges the typical assumption that class attendance is essential for high performance and may point to alternative learning strategies like self-study or asynchronous material review. For the model's time management feature, this supports allowing flexibility and emphasizing effective time use over rigid scheduling.

- **Gaming Frequency vs. High CGPA**
An especially unexpected finding is that both high and low gaming frequencies can lead to high CGPA, with perfect

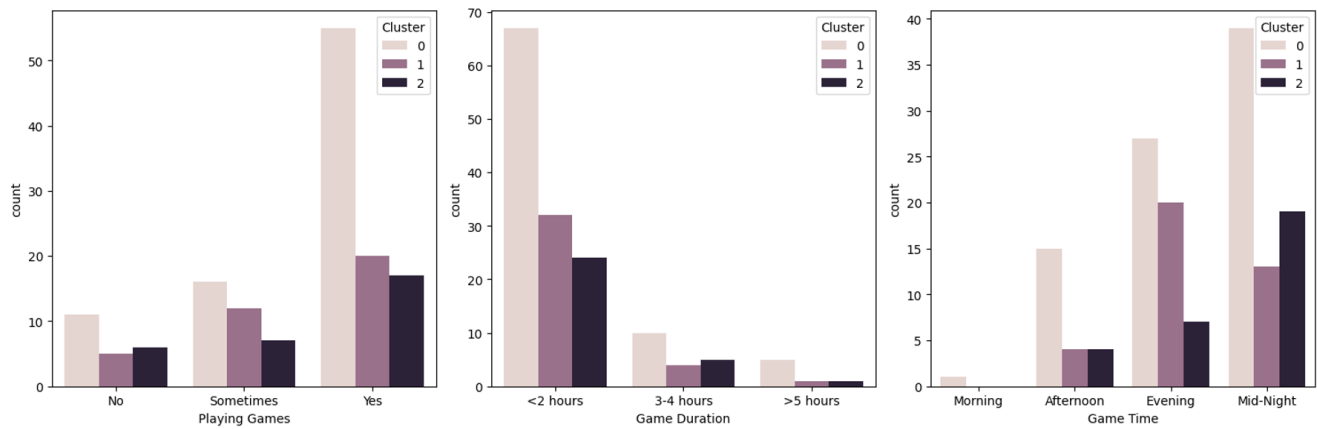


Figure 8: Distribution of Clustering Results Across Categorical Attributes (Parent Dataset: Playing Games, Game Duration, Game Time).

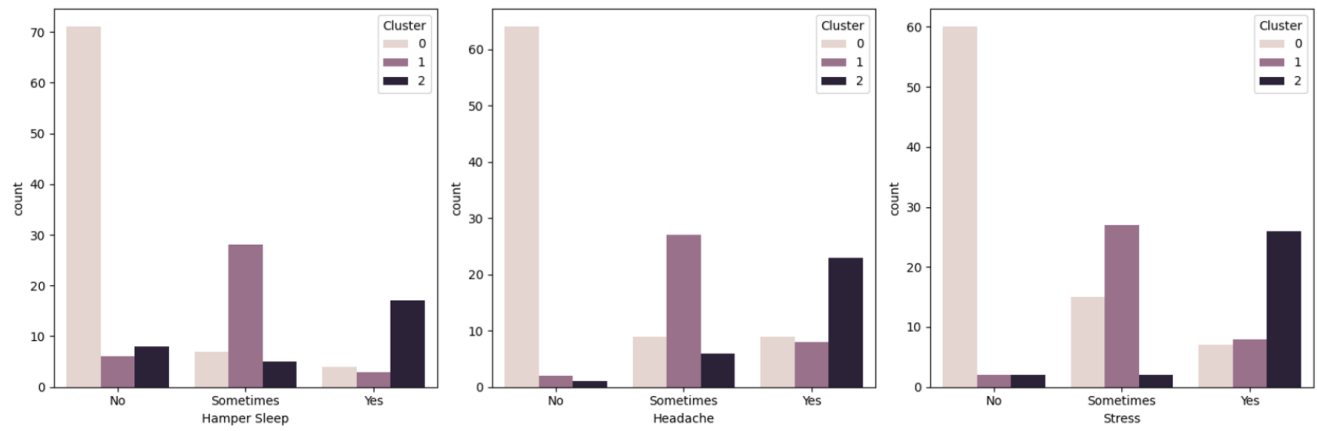


Figure 9: Distribution of Clustering Results Across Categorical Attributes (Parent Dataset: Hamper Sleep, Headache, Stress).

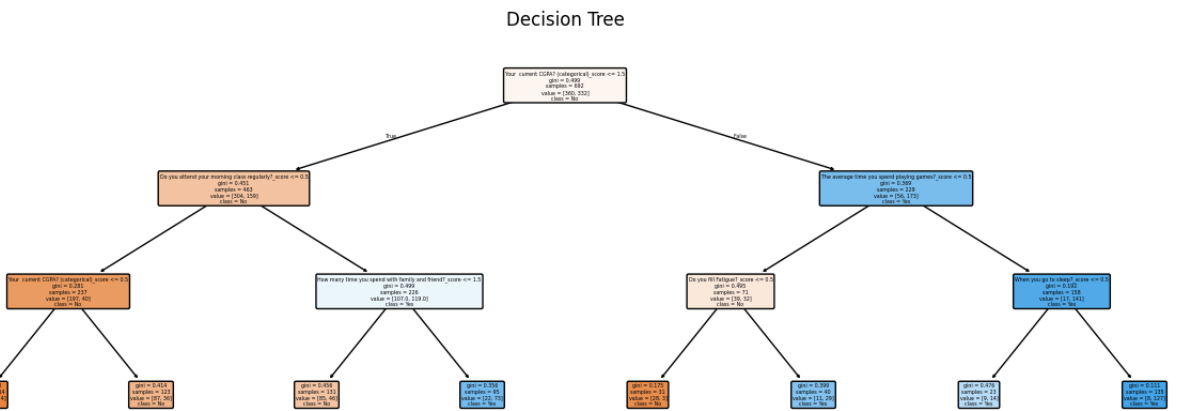


Figure 10: Decision Tree for Classifying Time Management

Out[52]:

	CGPA Category	Bed Time	Attend Morning Class	Avg Game Time	Social Time	Fatigue
0	Average	Average	Yes	Moderate	Moderate	Yes
1	Low	Late	No	Low	Low	No
2	High	Average	No	High	Moderate	No
3	High	Healthy	Yes	Moderate	Low	No

Figure 11: Cluster Centroids (K = 4) for Time Management Dataset.

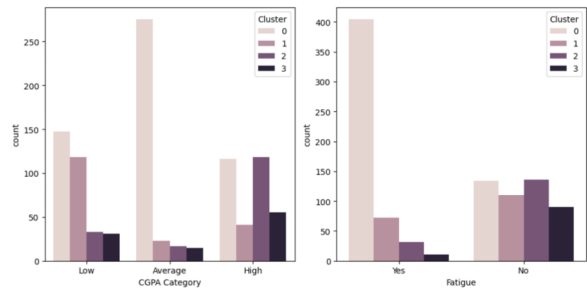


Figure 12: Distribution of Clustering Results Across Categorical Attributes (Time Management: CGPA Category, Fatigue).

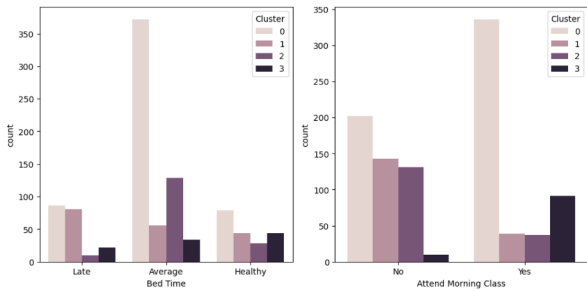


Figure 13: Distribution of Clustering Results Across Categorical Attributes (Time Management: Bed Time, Attend Morning Class).

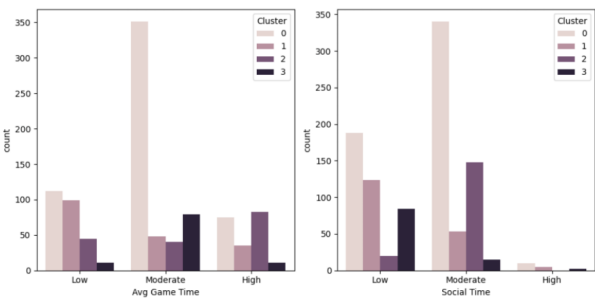


Figure 14: Distribution of Clustering Results Across Categorical Attributes (Time Management: Avg Game Time, Social Time).

confidence and high lift (2.99). This duality suggests that gaming itself is not necessarily harmful if balanced properly. For high performers, gaming may serve as a stress relief tool, while those who game less may allocate more time for academics. The bot can use this nuance to avoid blanket anti-gaming suggestions and instead promote healthy balance and reflection on how gaming fits into one's routine.

• **Moderate Gaming and Top CGPA**

Reinforcing the previous finding, students who spend a moderate amount of time gaming are also likely to report a perfect CGPA (4.0). This suggests that not only is moderate gaming not detrimental, but it might correlate with structured time management and cognitive recovery during breaks. The model could use this insight to reinforce moderate entertainment habits as part of a healthy academic lifestyle. However, when compared to low gaming habit students, it is found that the students with moderate gaming habits, with 4.0 CGPA, have a higher support (0.043 support, around 43 students) than students with 4.0 CGPA and low gaming habits (0.02 support, around 20 students). This shows a significant effect of a higher academic performance when involved in a moderate gaming habit.

• **Social Time vs. High CGPA**

Surprisingly, students with low (support 0.017) or moderate (support 0.057) time spent with family and friends are often associated with high CGPA. While excessive social interaction might interfere with study schedules, moderate social engagement appears to strike a balance that supports academic focus without complete isolation. This insight challenges the notion that higher social involvement always yields better outcomes, instead suggesting that intentional and limited socialization supports academic goals.

4.3 Mental Health

4.3.1 Classification.

In Figure 15, the root node splits on **Suicidal Thoughts Score** at a threshold of 0.5. This feature is the most influential predictor. Students with higher suicidal ideation (score > 0.5) are overwhelmingly classified as depressed (class = 1), with a large majority (4840 out of 9681 total samples) falling in this category.

Left Subtree (Lower Suicidal Thoughts Score). The tree further splits on **Financial Stress Score** ≤ 1.5 . Most samples here are classified as not depressed (class = 0), showing the protective role of low financial stress. This path continues to consider **Academic Pressure Score**: The leaf node with the lowest Gini (0.137) shows high purity: 1113 students, almost all not depressed. Another branch (Gini = 0.325) with 1189 samples is also mostly not depressed. The rightmost node under this path (Gini = 0.487) is more ambiguous, containing a mix of both classes.

Right Subtree (Higher Suicidal Thoughts Score). The split on **Financial Stress Score** ≤ 0.5 indicates that even with high suicidal thoughts, financial stress plays a strong role in determining depression outcomes. Academic Pressure Score remains a key downstream factor: One of the purest leaf nodes (Gini = 0.219, 3285 samples) shows strong evidence of depression when academic pressure is

high and financial stress is low. Another node with 1695 samples and Gini = 0.408 also shows high likelihood of depression. A slightly mixed node (Gini = 0.448) still leans toward depression classification.

Insights.

- **Suicidal ideation** is the most powerful indicator of depression in this model.
- **Financial stress and academic pressure** both play subtle but consistent roles in modulating mental health risk.
- **High academic pressure** is especially important in distinguishing ambiguous cases, especially when suicidal thoughts and financial stress do not strongly indicate depression.

Limitations.

- A few nodes (e.g., Gini ≈ 0.48) indicate classification uncertainty and may benefit from additional features or ensemble models.
- The data relies on self-reported scores, which could introduce bias.
- The large class imbalance (e.g., many more depressed cases in one path) may skew decision boundaries.

4.3.2 Clustering.

Clustering analysis of the Mental Health dataset was performed, resulting in three distinct clusters ($K = 3$) as determined by the distribution of centroids across key academic, lifestyle, and mental health attributes. The representative centroids for each cluster, summarized in Table 16, reveal clear differences in academic performance, psychological well-being, and lifestyle factors among student groups.

- **Cluster 0:** Identified as 'High-Risk Stressed Achievers', is characterized low academic achievement and high academic pressure, coupled with poor study satisfaction and a high prevalence of suicidal thoughts and depression. Despite moderate financial stress and no notable family history of mental health issues, these students are particularly vulnerable, highlighting the critical impact of academic stress on mental well-being.
- **Cluster 1:** Labeled as 'Mentally Healthy High Performers', comprises students who excel academically while maintaining low academic pressure and high study satisfaction. This group reports low financial stress and minimal mental health concerns, including an absence of suicidal ideation and depression, even though many have a family history of mental health issues. Their profile underscores the protective effect of balanced academic demands and satisfaction on overall mental health.
- **Cluster 2:** Referred to as 'Moderate Performers with Financial Stress', includes students with average academic results and high study satisfaction, but who face significant financial stress and a family history of mental health issues. Despite healthy lifestyle habits, this group shows elevated rates of depression and suicidal thoughts, suggesting that financial and familial factors can undermine mental health even among those with otherwise positive routines.

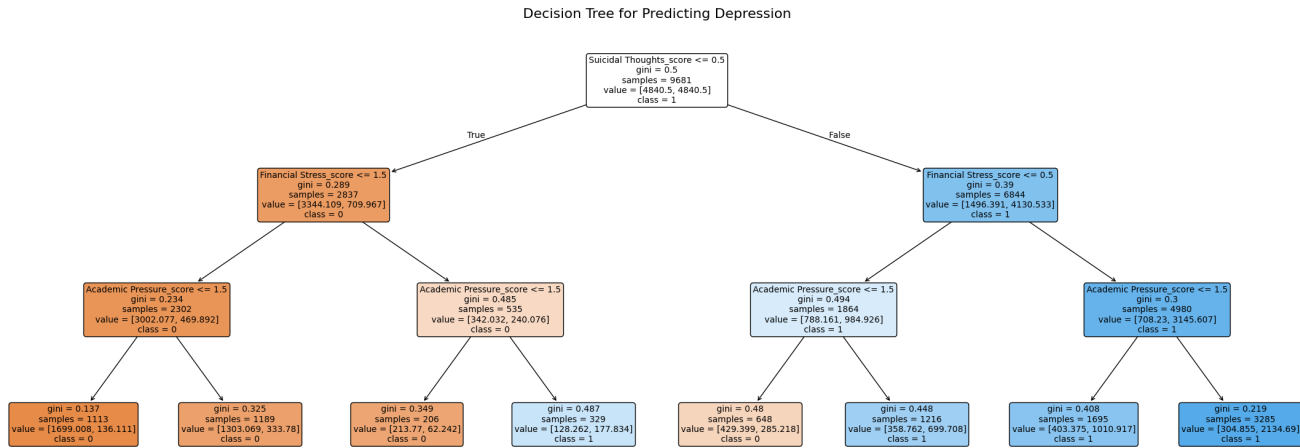


Figure 15: Decision Tree for Classifying Mental Health

Figures 17 to 19 demonstrate that academic, lifestyle, and psychosocial factors are closely linked, with attributes such as academic pressure, study satisfaction, sleep, and financial stress clustering in distinct patterns. High academic pressure often coincides with unhealthy habits and poor mental health, while students with high achievement and satisfaction generally report healthier routines. More importantly, elevated financial stress and a family history of mental health issues are strongly associated with depression and suicidal thoughts, even among those with positive lifestyle behaviors. These findings highlight that student well-being is shaped by the combined effects of academic, behavioral, and socioeconomic influences, emphasizing the need for support strategies integrated in our DiTTO study companion proposal.

4.3.3 Pattern Mining.

- **Academic Pressure = Low, Financial Stress = Low, Suicidal Thoughts = Yes, Work Study Hours = Low → Depression = 0**

This pattern reveals an unexpected insight: even when a student reports suicidal thoughts, the presence of low academic pressure, low financial stress, and low work-study burden is associated with a non-depressed (Depression=0) status, with a confidence of 82% and support 0.014. While suicidal thoughts typically signal mental health risk, this rule suggests that not all such cases are captured as depression in the data. This could indicate either underdiagnosis or a more complex emotional landscape where suicidal ideation exists without meeting the full criteria of depression. The bot should treat such combinations cautiously and potentially escalate for deeper checks.

- **Academic Pressure = High, Dietary Habits = Healthy, Financial Stress = High, Suicidal Thoughts = Yes → Depression = 1**

This rule highlights a strong correlation between high academic and financial stress coupled with suicidal thoughts, even when the individual maintains healthy eating habits.

With over 94% confidence and support 0.018, it confirms that mental well-being can deteriorate despite visible healthy routines. The presence of suicidal ideation and stress appears to outweigh protective lifestyle factors like nutrition. For our Discord extension model, this underlines the need to give greater weight to emotional and cognitive indicators over physical habits when assessing mental health.

- **CGPA = High, Financial Stress = Moderate, Study Satisfaction = High, Suicidal Thoughts = Yes → Depression = 1**

This pattern is particularly surprising: despite reporting high academic performance (CGPA) and high satisfaction with studies, a student with moderate financial stress and suicidal thoughts is still very likely to be depressed, with nearly 78% confidence with support 0.022. It challenges the stereotype that high-achieving or seemingly content students are mentally resilient. This reinforces the importance of including direct mental health cues in any diagnostic process — even top performers may be struggling internally. The model should ensure these users are not overlooked and are provided support despite their outward success indicators.

5 Discussion and Conclusion

This project demonstrates the feasibility and value of integrating data science techniques into a practical system for supporting university students facing academic and mental health challenges related to gaming behavior. By analyzing structured survey responses and applying a weighted scoring system to key indicators—including gaming habits, sleep patterns, headache frequency, and psychological stress—we successfully classified students into two main groups: those struggling with time management and those experiencing mental health issues. Our proposed DiTTO Discord-based extension bot was designed to deliver personalized, accessible recommendations directly within the students' digital environment, enhancing convenience and engagement. To be specific, our DiTTO

Out[35]:

	CGPA (categorical)	Academic Pressure	Study Satisfaction	Sleep Duration	Dietary Habits	Suicidal Thoughts	Work Study Hours	Financial Stress	Family History	Depression
0	Low	High	Low	Very short	Unhealthy	Yes	High	Moderate	No	1
1	High	Low	High	Adequate	Moderate	No	Moderate	Low	Yes	0
2	Average	Moderate	High	Long	Healthy	Yes	Low	High	Yes	1

Figure 16: Cluster Centroids (K = 3) for Mental Health Dataset.

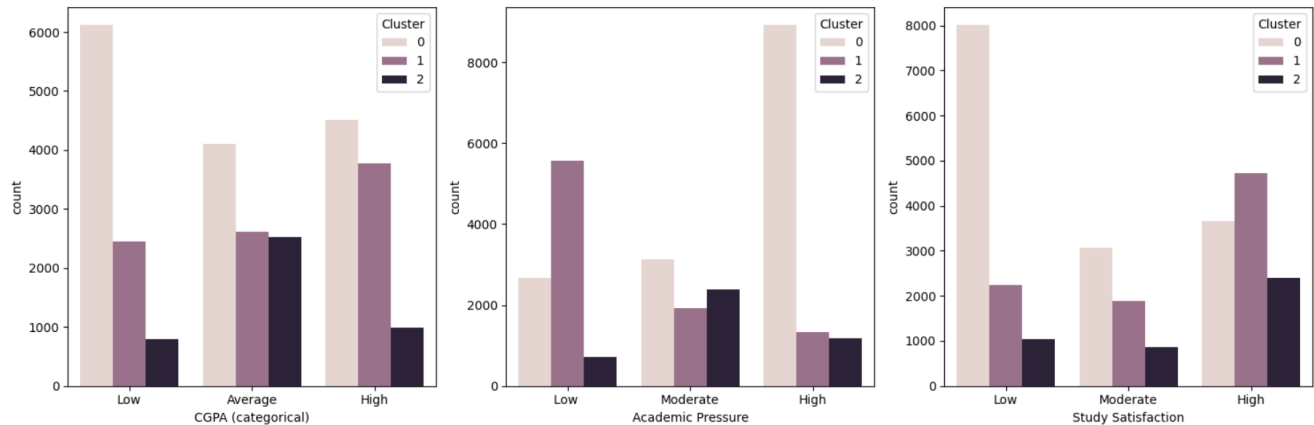


Figure 17: Distribution of Clustering Results Across Categorical Attributes (Mental Health: CGPA (Categorical), Academic Pressure, Study Satisfaction)

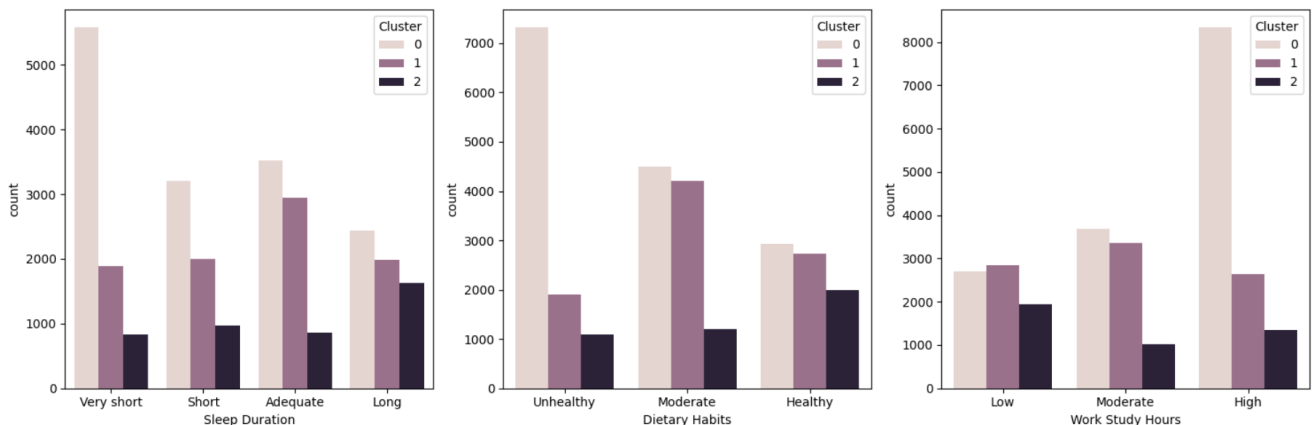


Figure 18: Distribution of Clustering Results Across Categorical Attributes (Mental Health: Sleep Duration, Dietary Habits, Work Study Hours)

extension will first analyze the student’s problem by asking questions and then deepen the question until the model can analyze the real problem that one is currently facing. From this point, DiTTo will be shaped into a suitable model to help the student overcome the problems that they had, and can also provide other functions, such as a time block, which can remind the students or lock the application if overused. DiTTo can also act as a virtual buddy for the students to overcome depression and give motivational words that could help students work on their mental health issues.

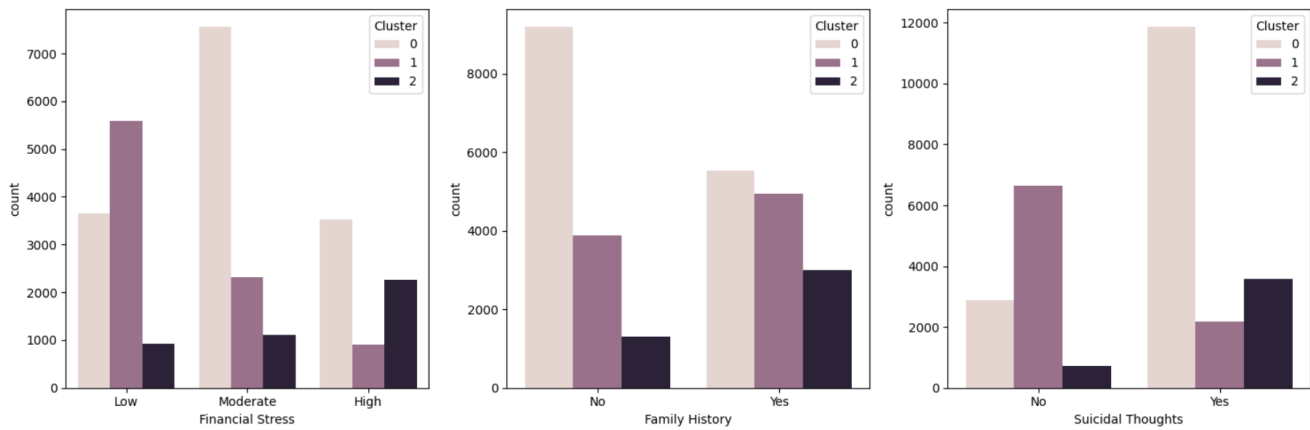


Figure 19: Distribution of Clustering Results Across Categorical Attributes (Mental Health: Financial Stress, Family History, Suicidal Thoughts)

Index	antecedents	consequents	support	confidence	lift
4266	Academic Pressure=High, Dietary Habits=Healthy, Financial Stress=High, Suicidal Thoughts=Yes	Depression=1	0.018141402552703283	0.9422718808193667	1.6092240570544807

Figure 20: Suprising Pattern That Shows Student Had Depression with Dietary Habits = Healthy

7911	CGPA (categorical)=High, Financial Stress=Moderate, Study Satisfaction=High, Suicidal Thoughts=Yes	Depression=1	0.02269467947798652	0.7747858017135864	1.323189173487347
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Figure 21: Suprising Pattern That Shows Student Had Depression with High CGPA and Financial Stress = Moderate

Beyond basic classification, the implementation of clustering algorithms enabled the identification of student subgroups with shared behavioral patterns, even when individuals did not fit neatly into a single label. This approach proved especially valuable for detecting early indicators of decline and for recognizing students at risk for both time management and mental health problems. Pattern mining further enriched the system's capabilities by uncovering frequent and significant associations in student responses, such as the co-occurrence of late-night gaming, sleep issues, and low academic performance. These insights empower DiTTo to adapt its feedback dynamically, offering proactive and data-driven support tailored to individual needs.

One of the primary challenges encountered was the clustering of predominantly categorical or ordinal data, which limited the applicability of traditional k-means algorithms. This was addressed by employing the K-Modes algorithm, optimized using the elbow method and robust initialization strategies, which enabled effective segmentation and the discovery of meaningful student groupings. The resulting clusters provided actionable profiles that informed more nuanced and targeted recommendations.

In summary, the integration of classification, clustering, and pattern mining within the DiTTo system resulted in a comprehensive understanding of students' academic and behavioral profiles. The project highlights the importance of combining multiple analytical techniques to address complex, real-world problems in student wellness. Future work will focus on deploying the DiTTo bot in

live student communities, evaluating its impact on academic performance and mental health outcomes, and further refining its recommendation logic through continuous data-driven feedback.

6 Team Contributions and Task Allocation

6.1 Individual Member Responsibilities

6.1.1 Ahmad Fahreen Qusyairi (2023320304).

- Contributed to the initial brainstorming and conceptual development of the project idea.
- Sourced relevant datasets suitable for the project objectives
- Prepared the slides and delivered the project proposal during the interim presentation.
- Performed data preprocessing and conducted analysis on the Parent dataset.
- Developed and contributed to the implementation of the classification model for the parent dataset through Python coding.
- Conducted pattern mining analysis for both child datasets: Time Management and Mental Health.
- Prepared and documented the pattern mining methodology and results in the final report (Methodologies and Evaluation).
- Contributed to the project discussion, final conclusions, and proposed ideas for future implementation of the Discord bot extension - DiTTo.

6.1.2 Hiew Long Chen (2023320156).

- Participated in the initial brainstorming and conceptualization of the project.
- Sourced relevant datasets suitable for the project objectives
- Prepared the slides for the interim presentation.
- Performed clustering analysis on all datasets—Parent, Time Management, and Mental Health
- Developed and contributed to the implementation of the clustering model for all datasets through Python coding.
- Documented the clustering methodology and findings in the final report (Methodologies and Evaluation).
- Conducted research to identify appropriate clustering techniques for the project.
- Contributed to the discussion and conclusions in the final report and proposed the future implementation idea of a Discord bot extension - DiTTo based on the project findings.

6.1.3 Nur Asha Syafiqah (2023320324).

- Contributed to the initial ideation and conceptual development of the project.
- Identified and selected appropriate datasets to support the project objectives.
- Designed and prepared the slides for the interim presentation.
- Researched and applied an unbiased classification approach, particularly for the Parent dataset, to ensure fair outcomes.
- Performed classification tasks across all datasets—Parent, Time Management, and Mental Health.
- Developed and implemented the classification models using Python for all datasets.
- Documented and explained the classification methodologies and evaluation sections in the final report.
- Finalized project report in Overleaf.
- Contributed to the refinement of the Discord bot extension concept for future implementation.

6.1.4 Nur Irdina (2023320029).

- Initiated the core project idea focusing on the relationship between gaming and student behavior, and contributed to its further development.
- Identified and sourced appropriate datasets relevant to the project goals.
- prepared and organized the interim presentation slides.
- Established the project's GitHub repository to support collaborative coding, documentation, and version tracking.
- Carried out complete data preprocessing and analysis for all datasets—Parent, Time Management, and Mental Health.
- Categorized and structured the data to support downstream tasks in clustering, classification, and pattern mining.
- Authored the introduction and data preprocessing methodology sections in the final report.
- Finalized the Overleaf report and ensured consistency across all sections.
- Contributed to the conceptualization of the Discord bot extension as a potential future implementation.

References

- [1] Jason Brownlee. 2020. *Cost-Sensitive Decision Trees for Imbalanced Classification*. Machine Learning Mastery. <https://machinelearningmastery.com/cost-sensitive-decision-trees-for-imbalanced-classification/> Accessed: 2025-06-21.
- [2] Jiawei Han, Jian Pei, and Hanghang Tong. 2023. *Data Mining: Concepts and Techniques* (4th ed.) [Lecture Slides]. https://hanj.cs.illinois.edu/bk4/bk4_slidesindex.htm. Accessed: 2025-06-21.
- [3] Zhexue Huang. 1997. A fast clustering algorithm to cluster very large categorical data sets in data mining. *Dmkd* 3, 8 (1997), 34–39.
- [4] Zhexue Huang. 1998. Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data mining and knowledge discovery* 2, 3 (1998), 283–304.
- [5] Zhexue Huang et al. 1997. Clustering large data sets with mixed numeric and categorical values. In *Proceedings of the 1st pacific-asia conference on knowledge discovery and data mining (PAKDD)*. Citeseer, 21–34.
- [6] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, et al. 2013. *An introduction to statistical learning*. Vol. 112. Springer.
- [7] Max Kuhn and Kjell Johnson. 2019. *Feature engineering and selection: A practical approach for predictive models*. Chapman and Hall/CRC.
- [8] Avinash Navlani. 2023. Decision tree classification in python tutorial. *datacamp* 12, 2 (2023).
- [9] Dorian Pyle. 1999. *Data preparation for data mining*. Morgan Kaufmann.
- [10] scikit-learn developers. 2024. *sklearn.tree.DecisionTreeClassifier — scikit-learn 1.4.2 documentation*. scikit-learn. <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html> Accessed: 2025-06-21.
- [11] Ramakrishnan Srikant. 1996. *Fast algorithms for mining association rules and sequential patterns*. The University of Wisconsin-Madison.
- [12] Pang-Ning Tan, Michael Steinbach, and Vipin Kumar. 2016. *Introduction to data mining*. Pearson Education India.